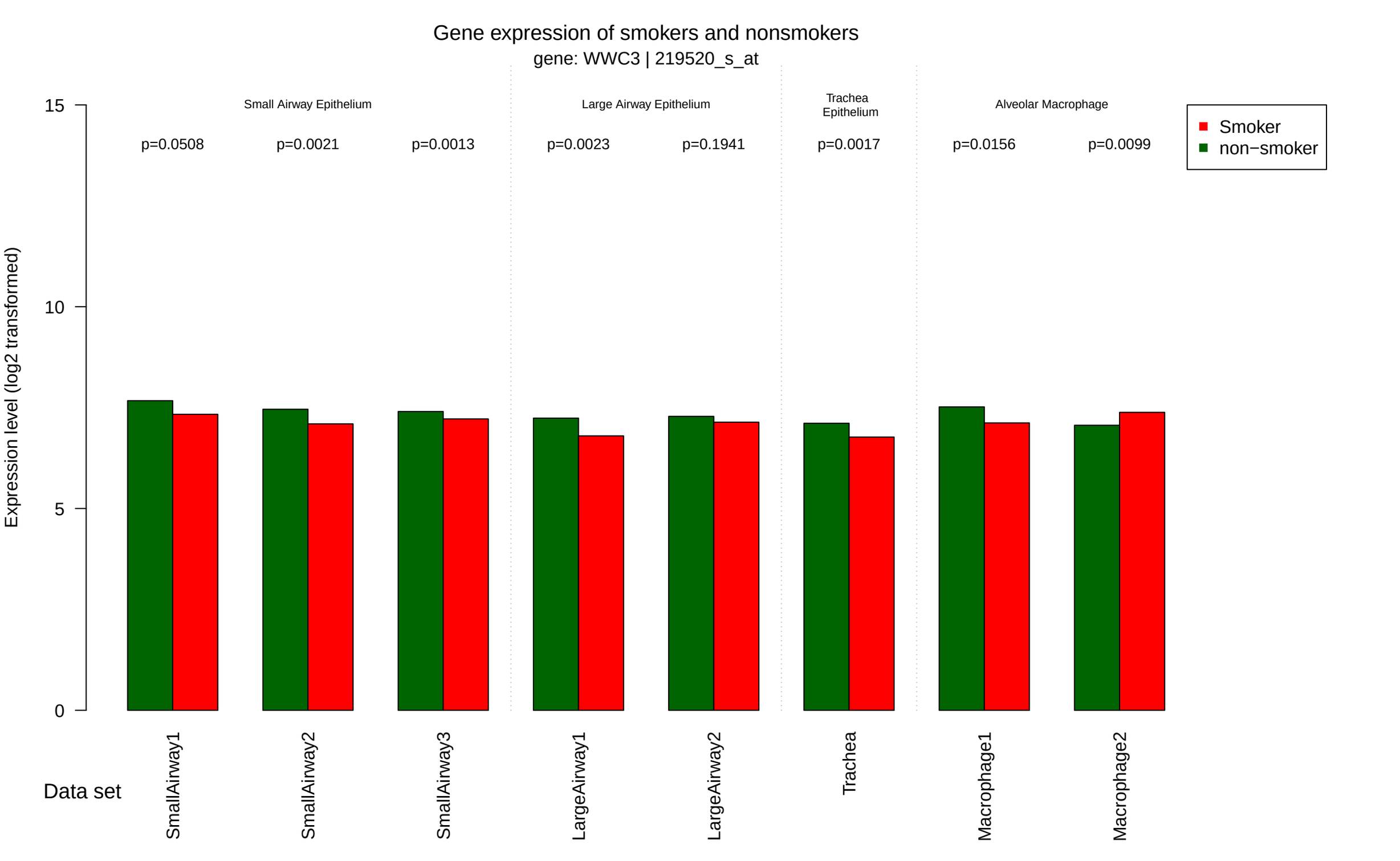
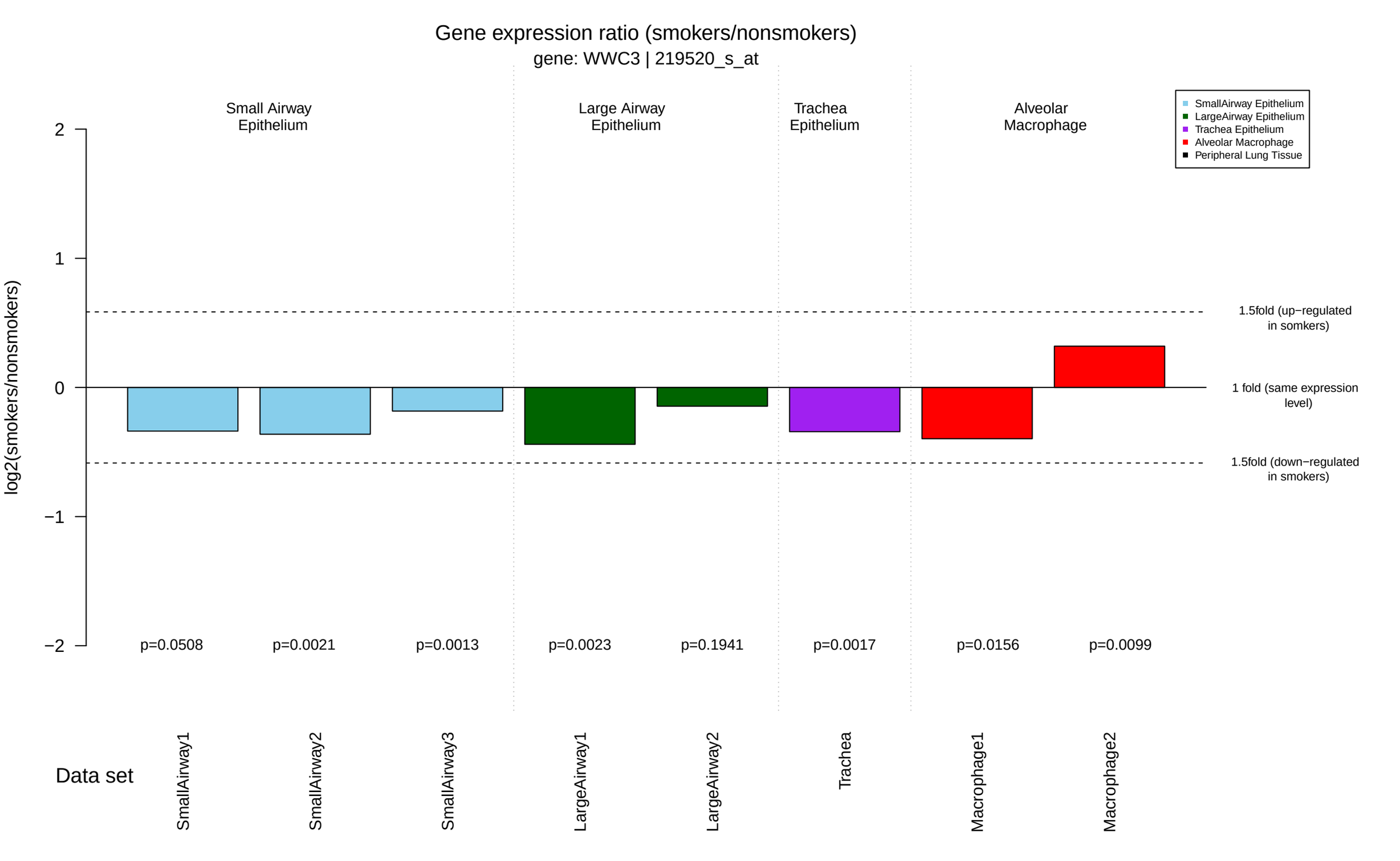
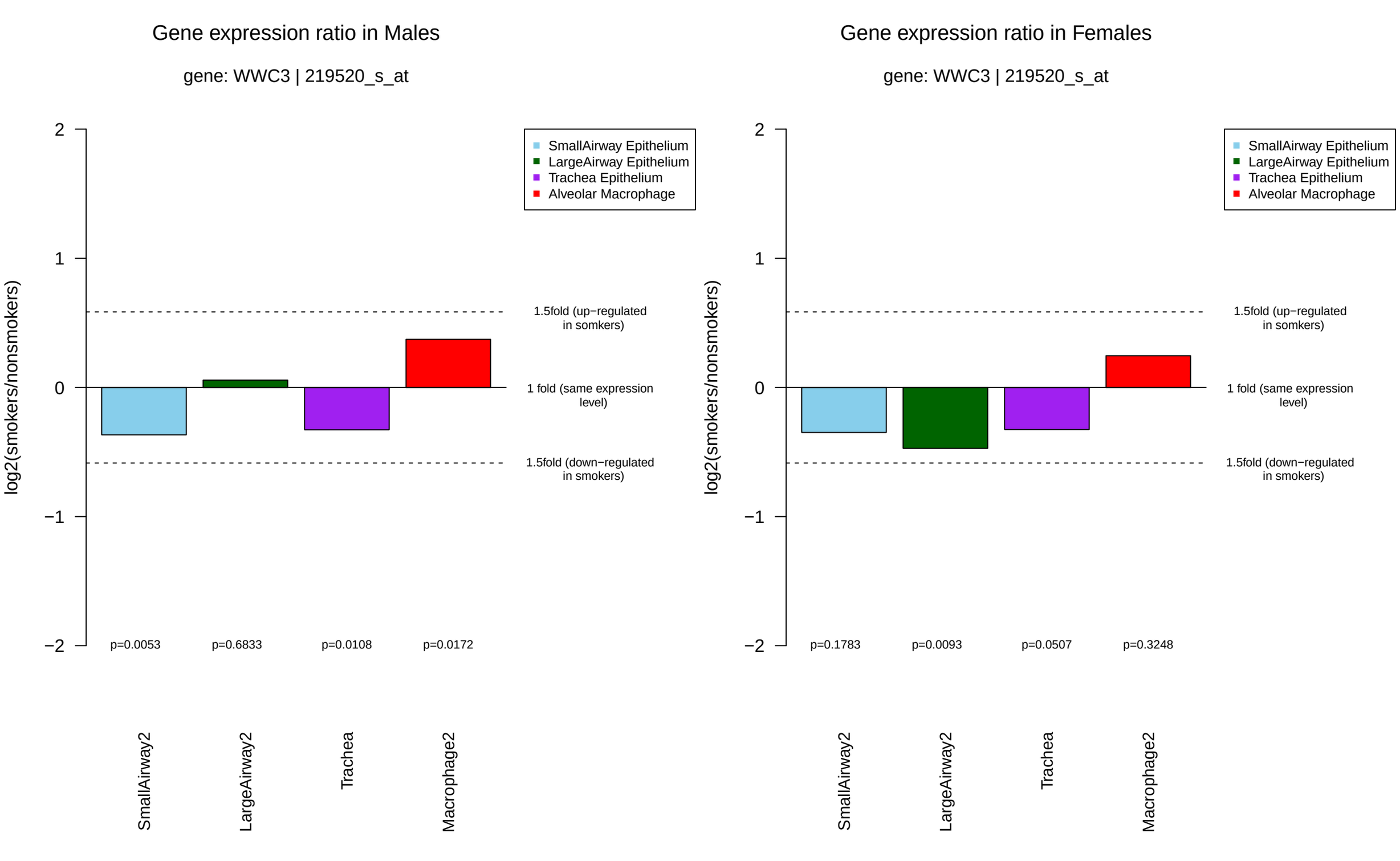
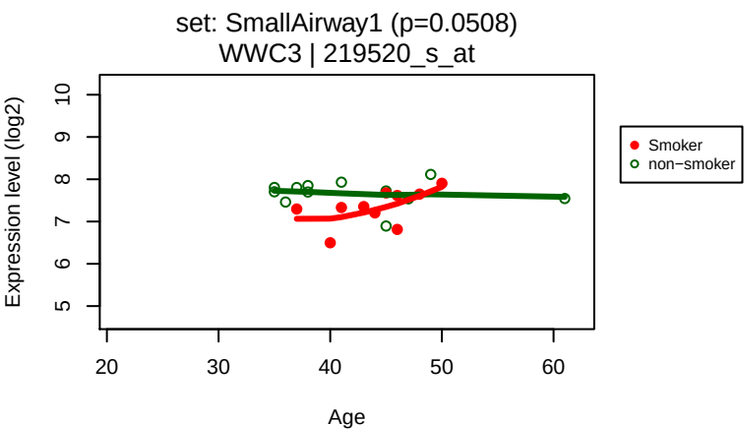
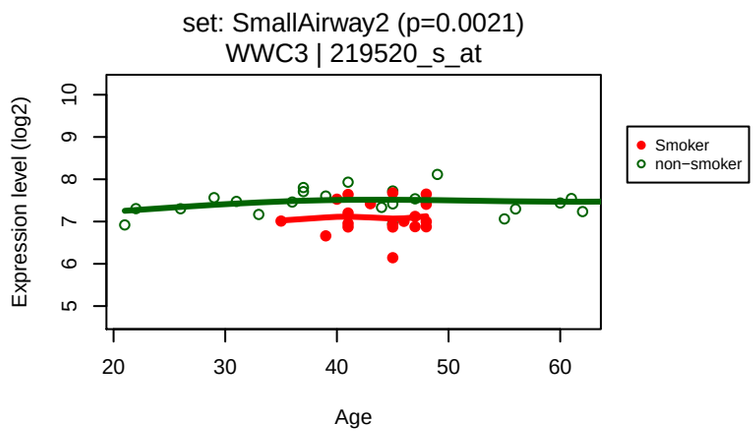
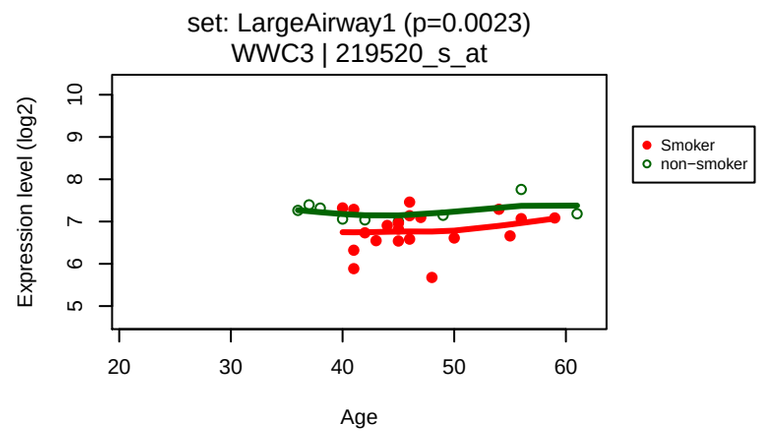

Gene expression of smokers and nonsmokers
gene: WWC3 | 219520_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0508
p=0.0021
p=0.0013
p=0.0023
p=0.1941
p=0.0017
p=0.0156
p=0.0099
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

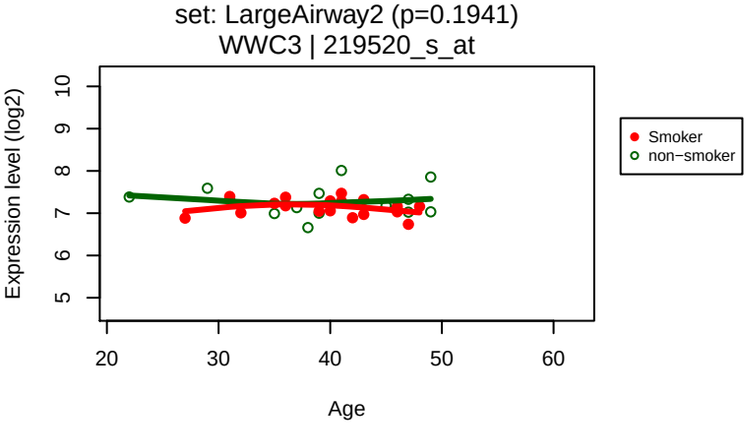

Gene expression ratio (smokers/nonsmokers)
gene: WWC3 | 219520_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0508
p=0.0021
p=0.0013
p=0.0023
p=0.1941
p=0.0017
p=0.0156
p=0.0099
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

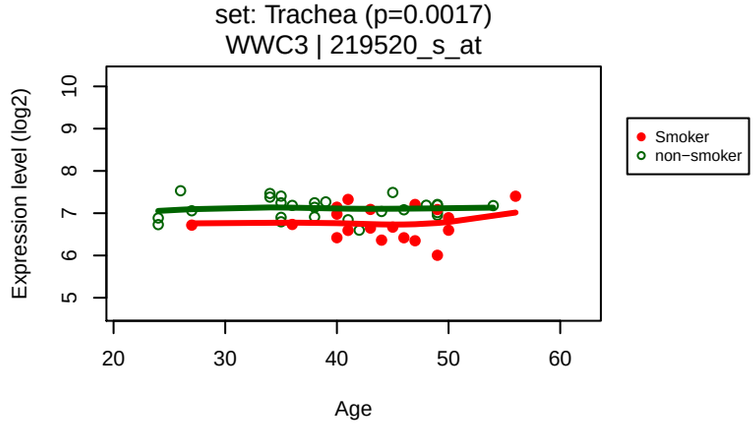

Gene expression ratio in Males
gene: WWC3 | 219520_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0053
p=0.6833
p=0.0108
p=0.0172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WWC3 | 219520_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1783
p=0.0093
p=0.0507
p=0.3248
SmallAirway2
LargeAirway2
Trachea
Macrophage2

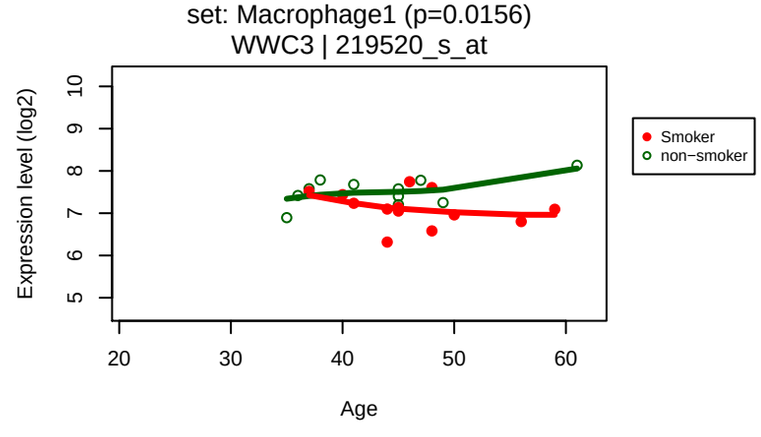

set: SmallAirway1 (p=0.0508)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker

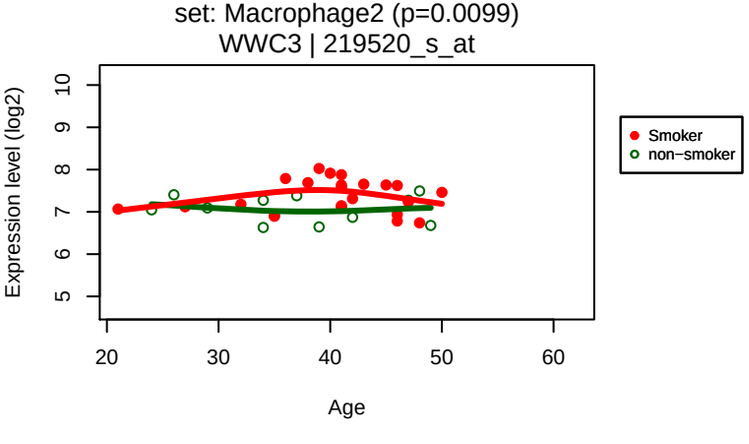

set: SmallAirway2 (p=0.0021)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker

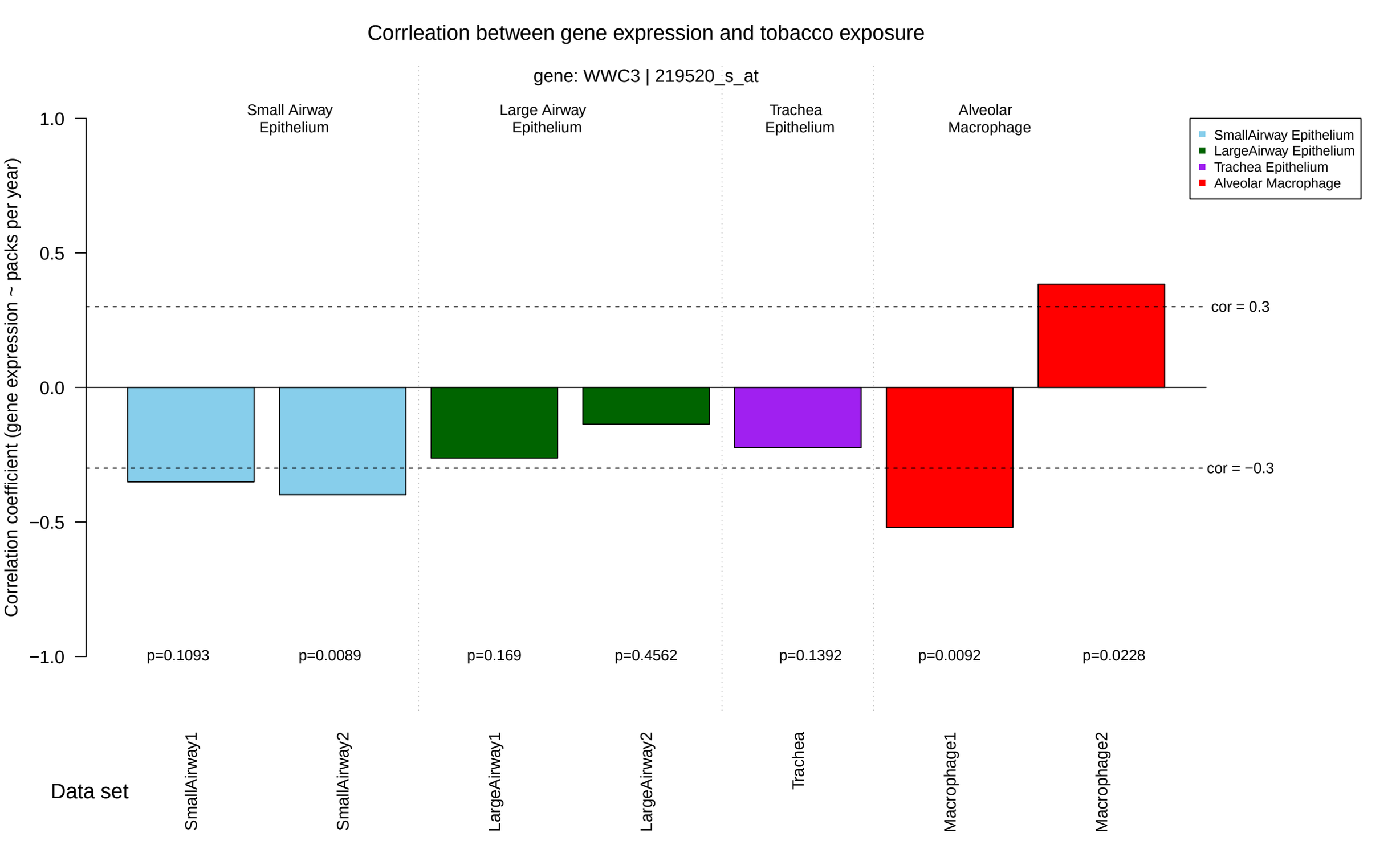

set: LargeAirway1 (p=0.0023)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1941)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0156)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0099)
WWC3 | 219520_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WWC3 | 219520_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1093
p=0.0089
p=0.169
p=0.4562
p=0.1392
p=0.0092
p=0.0228
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2